AJAY KUMAR PADALA

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SUMMARY

Data Scientist with experience applying analytics and AI in healthcare and life sciences to solve business problems, improve efficiency, and deliver insights. Experienced in translating complex data into practical solutions that support decision-making and create business value.

EXPERIENCE

Johnson & Johnson | *Data Scientist* — Remote | Jan 2025 – Present

- Developed interactive Tableau and Spotfire dashboards integrating multi-source clinical and operational data, enabling data-driven decision-making across development and commercial teams within the Innovative Medicine organization.
- Applied machine learning and statistical modeling techniques on high-dimensional pharmaceutical datasets, enabling faster identification of key trends in clinical trial outcomes and reducing manual analysis time by 25%.
- Conducted hands-on data discovery by writing complex SQL queries to extract, join, and structure data from laboratory and enterprise systems, reducing data preparation time by 30% for downstream analytics and reporting.
- Evaluated and prototyped Generative AI and LLM-based solutions for scientific document retrieval and data summarization, delivering a proof-of-concept that automated 40% of manual literature review workflows.

Johnson & Johnson | *Strategic Insights & Analytics Intern* — Boston, MA | Jun 2024 – Dec 2024

- Analyzed HIPAA-compliant healthcare and sales records, uncovering trends in patient outcomes and product adoption for leadership review.
- Designed interactive Tableau dashboards tracking product efficiency, supply chain operations, and business KPIs for executive briefings.
- Developed automated ELT workflows in Apache Airflow to integrate ERP and cloud data sources, reducing reporting cycles by 35%.
- Built forecasting models using regression and time-series methods, improving demand prediction and pricing accuracy by 20%.

Beats by Dre (Apple Inc.) | *Consumer Insights Analyst – Externship* — Boston, MA | Mar 2024 – May 2024

- Informed the Beats Solo 4 premium positioning strategy by analyzing consumer behavior data, identifying unmet demand for spatial audio features and recommending dynamic head-tracking integration to the product team.
- Designed an Apache Airflow pipeline architecture to reduce manual reporting by 40% and prototyped KPI dashboards in Databricks and Looker Studio that consolidated five weekly reports into a single live view for marketing and product leads.

Data Minds Analytics | *Data Analyst* — Hyderabad, India | Nov 2021 – Nov 2022

- Retained over $2M in annual revenue for a hospitality chain by engineering customer segmentation and churn prediction models in Python that reduced the annual customer churn rate by 14% within the first engagement quarter.
- Strengthened risk detection across a $50M+ lending portfolio by 16% by developing credit-risk scoring models in SAS for a major banking client, incorporating advanced feature engineering and rigorous model validation techniques.

SKILLS

- **Languages:** Python (Pandas, NumPy, PySpark, scikit-learn, XGBoost, TensorFlow, PyTorch, Matplotlib, Seaborn), R, SQL
- **Machine Learning:** Classification, Regression, NLP, Time-Series Forecasting, Deep Learning (CNNs, Transformers), RAG, LLM Applications
- **Data & Cloud:** Snowflake, Databricks, Apache Spark, PostgreSQL, AWS (S3, Redshift), Azure ML, dbt, Docker
- **Tools & Analytics:** Tableau, Spotfire, Apache Airflow, Power BI, Looker Studio, MLflow, Excel (Advanced), Git, JIRA, Confluence

ACADEMIC PROJECTS

HRV-Based Stress Classification App

- Built a Django web application with a 1D-CNN model and ANOVA F-test feature selection to classify four stress levels from heart rate variability signals, achieving 92% validation accuracy on the WESAD dataset and deploying a real-time inference endpoint for researcher use.

Intelligent Document Q&A Chatbot

- Developed a RAG-powered Q&A system using GPT-4, LangChain, and Chroma vector DB that achieved 90% answer accuracy on domain-specific document queries, deployed via a Streamlit interface with session memory and source citation.

EDUCATION

University of Massachusetts, Dartmouth, College of Arts and Sciences — Jan 2023 – Dec 2024
Master of Science, Data Science (GPA: 3.7/4.0)

CVR College of Engineering, Hyderabad, India — Jul 2018 – May 2022
Bachelor of Engineering, Electrical Engineering (GPA: 7.5/10)

CERTIFICATIONS & LEADERSHIP

- AWS Certified Cloud Practitioner – Amazon Web Services, 2025
- HIPAA Awareness Training Certification – American Health Training, 2025
- Google Data Analytics Professional Certificate – Google, 2025
- Oracle Cloud Infrastructure Generative AI Professional & AI Foundations Associate – Oracle, 2025
- Teaching Assistant, Artificial Intelligence – University of Massachusetts, Dartmouth